3 Complex Analysis 2024
Tutorial 11

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Complex Logarithms

Problem 44 Calculate

$$\text{(a)}\ \log(-i), \quad \text{(b)}\ \log\left(\frac{1}{2}(1+\sqrt{3}i)\right), \quad \text{(c)}\ i^{\frac{1}{2}}, \quad \text{(d)}\ i^{i+1}, \quad \text{(e)}\ (i+1)^i.$$

Problem 45 Give an example that shows that, in general,

$$\mathrm{Log}(zw) \neq \mathrm{Log}(z) + \mathrm{Log}(w) \quad (z, w \in \mathbb{C}).$$

Rouche's Theorem

Problem 46 For each of the following polynomials p, determine how many zeros p has in the unit disc $\{z \in \mathbb{C} : |z| < 1\}$:

(a) $p(z) = z^7 - 5z^4 + z^2 - 2$,

(b) $p(z) = 2z^5 - z^3 + 3z^2 - z + 8$,

(c) $p(z) = z^9 - 2z^6 + z^2 - 8z - 2$.

Problem 47 How many zeros does the polynomial $p(z) = z^4 - 5z + 1$ has in the annulus $\{z \in \mathbb{C} : 1 < |z| < 2\}$?

Problem 48 Let f be a holomorphic function in the closed unit disc $\overline{D}(0;1) = \{z \in \mathbb{C} : |z| \leq 1\}$. Assume that $|f(z)| < 1$ for all z with $|z| = 1$. How many solutions does the equation $f(z) = z$ has in the open unit disc $D(0;1) = \{z \in \mathbb{C} : |z| < 1\}$?

Problem 49 (a) Show that for any $\lambda > 1$ the equation

$$z + e^{-z} = \lambda$$

has exactly one solution in the right half-plane $\{z \in \mathbb{C} : \mathrm{Re}\, z > 0\}$.

(b) Prove that this solution is real.